

RW BLE iOS SDK User Guide

1. Introduction

This document explains the functional APIs and usage scenarios provided by the SDK.

This document applies only to RW company Bluetooth devices.

1.1 Supported Platforms and Languages

- iOS 12 and above, Objective-C language.

1.2 Terminology

- App: Refers to the application running on a mobile phone or tablet.
- Device: Refers to wearable hardware devices, such as watches and rings.
- Upload: Refers to data sent from the device to the App.
- Download: Refers to data sent from the App to the device.

1.3 Notes

1. This document uses Objective-C for all examples. If you use Swift, you must import the corresponding Objective-C header files in the project's Bridging Header.
2. The SDK does not provide a simulator version, because BLE cannot be debugged on the simulator, and some third-party libraries used by the SDK do not support the simulator environment.

2. Quick Start

Step 1: Install the latest version of Xcode

To develop with the RW BLE iOS SDK, Xcode must be installed.

Step 2: Manually add dependencies

Manually add `DHBluetooth.framework` to your project.

Frameworks, Libraries, and Embedded Content

Name	Embed
 DHBluetooth.framework	Do Not Embed
 DHFoundation.framework	Embed & Sign
 DHUIKit.framework	Embed & Sign

Step 3: Configure info.plist

```
//Add Bluetooth usage descriptions to info.plist.  
NSBluetoothAlwaysUsageDescription  
NSBluetoothPeripheralUsageDescription
```

Step 4: Initialize the SDK

```
//Initialize the DHBLESDK in AppDelegate.  
- (void)initBLESDK{  
    [DHBLECentralManager setLogStatus:YES];  
    [DHBLECentralManager initWithServiceUids:@[]];  
}
```

⚠ Caution

When logging is enabled `[DHBLECentralManager setLogStatus:YES]`, log files will be stored in the `Document/DeviceLog` directory.

3. API Reference)

3.1 Device Scanning, Connection, Binding, and Reconnection

3.1.1 Scan for Devices

Description: Call `startScan` to scan for BLE devices and implement the `DHBLEConnectDelegate`.

If the returned `DHPeripheralModel` has an empty `macAddr`, it indicates that the device is already paired in system settings.

```
// 1. 开始搜索  
[DHBLECentralManager startScan];  
  
//2. 设备委托  
[DHBLECentralManager sharedInstance].connectDelegate = self;  
  
//3. DHBLEConnectDelegate接口会回调搜索到的蓝牙设备  
- (void)centralManagerDidDiscoverPeripheral:(NSArray <DHPeripheralModel  
*>*)peripherals
```

3.1.2 Stop Scanning

Description: Stop scanning for BLE devices.

```
[DHBLECentralManager stopScan];
```

3.1.3 Connect Device

Description: Connect to a specified device and set the connection delegate.

```
// 1. Initialize and register the callback.
[DHBLECentralManager sharedInstance].connectDelegate = self;

// 2. Connect device
DHPeripheralModel *deviceModel = self.deviceArray[indexPath.row];
[DHBLECentralManager connectDeviceWithModel:deviceModel];

// 3. Implement and receive callbacks for Bluetooth connection status.
@protocol DHBLEConnectDelegate <NSObject>
```

`DHBLEConnectDelegate` Interface Description:

method	illustrate
<code>centralManagerDidDiscoverPeripheral</code>	After calling <code>startScan</code> , the system will return a callback when a device is found.
<code>centralManagerDidConnectPeripheral</code>	After calling <code>connectDeviceWithModel</code> , the function will return upon successful connection.
<code>centralManagerDidFunctionMenu</code>	The function will return after successfully obtaining the device configuration table; business operations should be performed after this point.
<code>centralManagerDidDisconnectPeripheral</code>	A callback function will be triggered when the Bluetooth connection is disconnected.
<code>centralManagerDidFailedPeripheral</code>	Bluetooth failure will trigger a callback.
<code>centralManagerDidUpdateState</code>	The Bluetooth switch state change will trigger a callback.

Tip

After connecting, business operations should only be performed after the `centralManagerDidFunctionMenu` method has been called.

(1) Listening to `BluetoothNotificationConnectStateChange` can also provide information about changes in the connection state.

```
[[NSNotificationCenter defaultCenter] addObserver:self
selector:@selector(connectStateChange:)
name:BluetoothNotificationConnectStateChange object:nil];

- (void)connectStateChange:(NSNotification *)ntf
{
    NSLog(@"NewHomeController connectStateChange");
    if ([DHBluetoothManager sharedInstance].isConnected){
        self.infoDeviceStateLb.text = @"Connected";
    }
    else{
        self.infoDeviceStateLb.text = @"Disconnected";
    }
}
```

(2) `[DHBLECentralManager isConnected]` can be used to check if the connection was successful.

3.1.4 Disconnect the device.

Interface description: Disconnect the currently connected device;

```
[DHBLECentralManager disconnectDevice];
```

3.1.5 Binding and automatic reconnection, unbinding

3.1.5.1 DHBLECentralManager Save to the current device locally.

After setting this, the current device's UDID will be saved, and it will automatically reconnect.

Method Description:

```
+(void)setBindedStatus:(BOOL)isBinded
```

Example of usage:

```
//Save locally, and reopening the file will reconnect.  
[DHBLECentralManager setBindedStatus:YES];
```

3.1.5.2 DHBLECentralManager Delete locally saved data

After this setting is applied, the UDID of the current device will be deleted, similar to unlinking the device locally.

Method Description:

```
+(void)setBindedStatus:(BOOL)isBinded
```

Example of usage:

```
[DHBLECentralManager setBindedStatus:NO];  
[DHBLECentralManager disconnectDevice];
```

3.1.6 Equipment Configuration Table

Important

Due to the variety of device models and their differing supported features, a feature table has been introduced to allow users to check the supported functions of each device. Please refer to the DeviceFuncV2Model class for details. The feature table content can be saved according to your specific business needs.

```
-(void)centralManagerDidFunctionMenu:(DeviceFuncV2Model *)deviceFuncModel
```

DeviceFuncV2Model class attribute definitions:

attribute	illustrate
isPushMsgEnableSwitch	Enable or disable message control switch
isAlarm	Does it support an alarm clock?
isBackLight	Does it support screen sleep time settings?
isSupportWorkout3	Does it support multiple sports modes?
isSupportMuslimCountSwitch	Does it support enabling/disabling Muslim prayers?
isSupportHrSpO2Alert	Does it support HR and SpO2 alarm notification functions?
isSupportMotoVibrationLevel	Does it support motor vibration alerts?
isDataTypeActivity	Does it support step counting?
isDataTypeHeart	Does it support heart rate monitoring?
isDataTypeBloodPressure	Does it support blood pressure measurement?
isDataTypeSleep	Does it support sleep mode?
isDataTypeSPO2	Does it support blood oxygen monitoring?
isDataTypeHRV	Does it support heart rate variability?
isDataTypeStress	Does it support pressure sensitivity?
isDataTypeBloodSugar	Does it support blood glucose monitoring?
isDataTypeMuslimCount	Do you support giving compliments/praise?

3.2 Device function operation

Error code definitions for API calls SendStateCode:

SendStateCode	value	illustrate
SendState_OK	0	OK
SendState_BLENoOpen	1	Failure, Bluetooth is not enabled.
SendState_BLEDisconnect	2	Failure, Bluetooth not connected.
SendState_TimeOut	3	Failure, Timeout
SendState_Failed	4	fail
SendState_DuplicateSend	5	Repeat command
SendState_NotSupport	6	Device not supported

3.2.1 Basic function command interface

3.2.1.0 Get SDK Version

Get the SDK version number.

Method Description:

[DHBluetoothCommand getSDKVersion]

Example of usage:

```
NSLog(@"%@", [DHBluetoothCommand getSDKVersion]);
```

3.2.1.1 Get Bluetooth MAC address

Because the iOS system cannot obtain the MAC address from the broadcast packet after pairing, this method is provided to retrieve it.

Method Description:

+ (void)ringGetMacAddress:(void(^)(int code, id data))block

Return parameter description:

parameter	type	illustrate	
model	DHDeviceInfoModel	class	macAddr: Mac地址

Example of usage:

```
[DHBluetoothCommand ringGetMacAddress:^(int code, id _Nonnull data) {
    DHDeviceInfoModel *tDeviceInfoData = data;
    NSLog(@"mac: %@", tDeviceInfoData.macAddr);
}];
```

3.2.1.2 Set user information

User information settings are related to step count, calories burned, and distance. During device initialization, the default settings are: gender 1, age 18, height 170cm, and weight 65 kg.

Method Description:

+ (void)setUserInfo:(DHUserInfoSetModel *)model block:(void(^)(int code, id data))block

Parameter Description:

parameter	type		illustrate
model	DHUserInfoSetModel	class	gender: Gender (0. Female, 1. Male) Height: height in cm, floating-point number Weight: weight in kg, floating-point number Step Goal: step count target value, currently no functionality Age: age

Example of usage:

```
DHUserInfoSetModel *userInfoModel = [[DHUserInfoSetModel alloc] init];
userInfoModel.gender = 1;
userInfoModel.height = 170;
userInfoModel.weight = 600;
userInfoModel.stepGoal = 8000;
userInfoModel.age = 20;
[DHBLECommand setUserInfo:userInfoModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"set ok");
    }
}];
```

3.2.1.3 Get firmware information

Interface description: Retrieves the firmware model, firmware version number, and UI version number;

```
[DHBLECommand getFirmwareVersion:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"getFirmwareVersion OK");
        DHFirmwareVersionModel *model = data;
        NSLog(@"model %@ version %@ UI version %@", model.deviceModel,
model.firmwareVersion, model.uiVersion);
    }
}];
```

3.2.1.4 Get battery level

Interface description: App retrieves device battery level.

```
[DHBLECommand getBattery:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBatteryInfoModel *model = data;
        NSLog(@"getBattery OK battery %zd", model.battery);
    }
}];
```

3.2.1.5 Get and set video control switches.

Configure whether to enable ring gesture control for browsing videos; [This function requires pairing with a Bluetooth HID device.](#)

```
//Set video control switch
DHVideoHidSetModel *tModeSetModel = [[DHVideoHidSetModel alloc] init];
tModeSetModel.isOpen = YES;
[DHBleCommand setVideoHid:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setVideoHid OK");
    }
}];

// Get video control switch
[DHBleCommand getVideoHid:^(int code, id _Nonnull data) {
    if (code == 0){
        DHVideoHidSetModel *model = data;
        NSLog(@"getVideoHid OK isOpen %d", model.isOpen);
    }
}];
```

3.2.1.6 Get and set LED screen brightness.

Method Description:

```
+(void)getRingLEDLight:(void(^)(int code, id data))block;
```

```
+(void)setRingLEDLight:(DHLedLightSetModel *)model block:(void(^)(int code, id data))block;
```

Parameter Description:

parameter	type		illustrate
model	DHLedLightSetModel	class	isOpen: false means off, true means (Levels 1-3) Light Level: 1 (dim light), 2 (soft light), 3 (bright light)

Example of usage:

```
// Get the LED screen brightness level.
[DHBleCommand getRingLEDLight:^(int code, id _Nonnull data) {
    if (code == 0){
        DHLedLightSetModel *model = data;
        NSLog(@"getRingLEDLight OK 开关 %d", model.isOpen);
    }
}];

// Set the LED screen brightness.
DHLedLightSetModel *tModeSetModel = [[DHLedLightSetModel alloc] init];
tModeSetModel.isOpen = YES; //
tModeSetModel.lightLevel = 3; //1 (dim light), 2 (soft light), 3 (bright light)
[DHBleCommand setRingLEDLight:tModeSetModel block:^(int code, id _Nonnull data) {
    {
```



```

    if (code == 0){
        NSLog(@"setRingLEDLight OK");
    }
}];

```

3.2.1.7 Get and set the wearing position.

```

// 获取佩戴位置
[DHBLECommand getRingWearHand:^(int code, id _Nonnull data) {
    if (code == 0){
        NSInteger tWearHand = [data intValue]; //0左手 1右手
        NSLog(@"getRingWearHand OK 佩戴位置 %zd", tWearHand);
    }
}];

//设置佩戴位置
uint8_t tModeSetModel = 0; 0: Left hand 1: Right hand
[DHBLECommand setRingWearHand:tModeSetModel block:^(int code, id _Nonnull data)
{
    if (code == 0){
        NSLog(@"setRingWearHand OK");
    }
}];

```

3.2.1.8 Starting and stopping photo taking

After activating the camera function, the device can use gesture control to take photos with the customized camera app.

Configuration table attribute: `isTakePhoto`

The `BluetoothNotificationCameraTakePicture` device sends a notification to take a picture, and then takes the picture.

```

// When the app enters the camera interface, a value of 1 controls the device to
enter the corresponding interface, and a value of 0 controls the device to exit.
[DHBLECommand controlCamera:1 block:^(int code, id _Nonnull data) {

}];

// The monitoring device issued a command to take a picture.
[[NSNotificationCenter defaultCenter] addObserver:self
selector:@selector(cameraTakePictureNotification)
name:BluetoothNotificationCameraTakePicture object:nil];

- (void)cameraTakePictureNotification {
    // Take a photo
    NSLog(@"cameraTakePictureNotification Take a photo");
}

```

3.2.1.9 Find devices

After initiating the search function, the device's light or screen will turn on.

Method Description:

```
+(void)controlFindDeviceBegin:(void(^)(int code, id data))block;
```

Example of usage:

```
[DHBleCommand controlFindDeviceBegin:^(int code, id _Nonnull data) {  
  
}];
```

3.2.1.10 Turn off the device, restore factory settings.

Method Description:

```
+(void)controlDevice:(NSInteger)type block:(void(^)(int code, id data))block;
```

Parameter Description:

parameter	type		illustrate
type	NSInteger	整形	1: Turn off 2: restore factory

Example of usage:

```
// 1关机 2 恢复出厂  
[DHBleCommand controlDevice:1 block:^(int code, id _Nonnull data) {  
  
}];
```

3.2.1.11 Alarm

Configuration table attribute: isAlarm

3.2.1.11.1 Get the alarms that have been set.

Method Description:

```
+(void)getAlarms:(void(^)(int code, id data))block
```

Example of usage:

```
//获取设备里已保存的闹钟  
[DHBleCommand getAlarms:^(int code, id _Nonnull data) {  
    if (code == 0) {  
        NSArray *tAlarmList = data;  
        NSLog(@"getAlarms %zd", tAlarmList.count);  
    }  
}];
```

3.2.1.11.2 Set an alarm

The current protocol does not support modifying individual alarms. Any operation to switch on/off or delete a single alarm requires resending the entire alarm configuration.

Method Description:

```
+(void)setAlarms:(NSArray <DHAlarmSetModel *>*)alarms block:(void(^)(int code, id data))block;
```

Parameter Description:

parameter	type	illustrate	illustrate
alarms	List	List	isOpen: true Open/false Off repeats: IntArray(7) Sunday to Saturday, set the corresponding element to 1 for repetition. hour: start hour minute: start minute

Example of usage:

```
//设置闹钟
DHAlarmSetModel *tAlarm1 = [[DHAlarmSetModel alloc] init];
tAlarm1.hour = 07; //时
tAlarm1.minute = 00; //分
tAlarm1.isOpen = true; //开关
tAlarm1.repeats = @[0,0,0,0,0,0,1]; //重复周期,周六重复

DHAlarmSetModel *tAlarm2 = [[DHAlarmSetModel alloc] init];
tAlarm2.hour = 8;
tAlarm2.minute = 00;
tAlarm2.isOpen = true;
tAlarm2.repeats = @[]; //单次闹钟

[DHBLECommand setAlarms:@[tAlarm1, tAlarm2] block:^(int code, id _Nonnull data)
{
}];
```

3.2.1.11.3 Delete all alarms

By passing an empty array to the `setAlarms` parameter, you can delete all alarms.

```
[DHBLECommand setAlarms:@[] block:^(int code, id _Nonnull data) {
}];
```

3.2.1.12 Setting and retrieving the number of vibrations

Set the number of times the device vibrates;

Configuration table property: `isSupportMotoVibrationLevel`

Method Description:

```
+ (void) setRingMotorLevel: (NSInteger) motorLevel motorNum: (NSInteger) motorNum block:
(void(^)(int code, id data)) block;

+ (void) getRingMotorLevel: (void(^)(int code, id data)) block;
```

Parameter Description:

parameter	type	illustrate	illustrate
motorLevel	Int	整形	Vibration intensity: 0: Off, 1: Low, 2: Medium, 3: High; <i>This function is not defined and can be ignored</i>
motorNum	Int	整形	The number of vibrations can be set (0-6 times), with a default setting of 2 times. Setting it to 0 will disable vibration.

Example of usage:

```
//设置
[DHBluetooth setRingMotorLevel:1 motorNum:2 block:^(int code, id _Nonnull data)
{
    if (code == 0){
        NSLog(@"set ok");
    }
}]];

//获取
[DHBluetooth getRingMotorLevel:^(int code, id _Nonnull data) {
    if (code == 0){
        DHVibrationLevelModel *tVibrationModel = data;
        NSLog(@"getRingMotorLevel Level %d num %d", tVibrationModel.vibrationLevel,
tVibrationModel.vibrationNumber);
    }
}]];
```

3.2.1.13 Setting and retrieving screen sleep mode settings.

- Set screen sleep mode and time;
- Configuration table property: isBackLightSleepMode

Method Description:

```
+ (void) setDisplaySleepMode: (DHBrightTimeSetModel *) model block: (void(^)(int code,
id data)) block

+ (void) getDisplaySleepMode: (void(^)(int code, id data)) block
```

Parameter Description:

parameter	type	illustrate	illustrate
DHBrightTimeSetModel	class		sleepOpen: Switch on (YES) or off (NO) sleepStartHour: Start time (hour) sleepStartMin: Start time (minute) sleepEndHour: End time (hour) sleepEndMin: End time (minute)

Example of usage:

```
//设置
DHBrightTimeSetModel *sleepModel = [[DHBrightTimeSetModel alloc] init];
sleepModel.sleepOpen = YES;
sleepModel.sleepStartHour = 20;
sleepModel.sleepStartMin = 00;
sleepModel.sleepEndHour = 06;
sleepModel.sleepEndMin = 00;
[DHBleCommand setDisplaySleepMode:sleepModel block:^(int code, id _Nonnull data)
{

}];

//获取
[DHBleCommand getDisplaySleepMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBrightTimeSetModel *tModel = data;
        NSLog(@"getDisplaySleepMode sleepOpen %d sleepStartHour %d sleepStartMin %d",
tModel.sleepOpen, tModel.sleepStartHour, tModel.sleepEndMin);
    }
}];
```

3.2.1.14 Setting and retrieving notification push settings

Message push notification switch;

Configuration table attributes: `isPushMsg` and `pushMsgSwitchValue` determine which applications on the device are supported.

Method Description:

```
+(void)ringSetAncs:(DHAncsSetModel *)model block:(void(^)(int code, id data))block
+(void)ringGetAncs:(void(^)(int code, id data))block
```

Parameter Description:

parameter	type	illustrate	illustrate
DHAncsSetModel	class		See the definition of the DHAncsSetModel class.

Example of usage:

```

//设置
DHAnCsSetModel *tAnCsModel = [[DHAnCsSetModel alloc] init];
tAnCsModel.isSMS = YES;
[DHBLECommand ringSetAnCs:tAnCsModel block:^(int code, id _Nonnull data) {

}];

//获取
[DHBLECommand ringGetAnCs:^(int code, id _Nonnull data) {
    if (code == 0){
        DHAnCsSetModel *anCsModel = data;

    }
}];

//按下面的顺序确定当前设备支持哪些应用消息。
UInt8 tBitRow = i;
if (i == 24){ //其它
    tBitRow = 0;
}
if (i > 0 && (tMsgSwitchValue & (1 << tBitRow)) < 1){ //不支持的消息全设置为false,不管从设备读到的值。
    continue;
}

self.msgOpenFlagArr = [NSMutableArray arrayWithArray:@[
    @(self.model.isCall),
    @(self.model.isSMS),
    @(self.model.isEmail),
    @(self.model.isSkype),
    @(self.model.isFacebook),
    @(self.model.isWhatsapp),
    @(self.model.isLine),
    @(self.model.isInstagram),
    @(self.model.isKakaotalk),
    @(self.model.isGmail),
    @(self.model.isTwitter),
    @(self.model.isLinkedin),
    @(self.model.isJLSinaWeiBo),

    @(self.model.isQQ),
    @(self.model.isWechat),
    @(self.model.isJLBand),
    @(self.model.isJLTelegram),
    @(self.model.isJLBetween),
    @(self.model.isJLNavercafe),
    @(self.model.isYoutube),
    @(self.model.isJLNetflix), //(1<<21)
    @(self.model.isMax), // (1<<22)
    @(self.model.isVkim), // (1<<23)
    //          @(self.model.isMessenger),

    @(self.model.isOther)
]];

```

3.2.1.14 Get and set whether the "likes" feature is enabled.

Set whether the "like" function is enabled;
Configuration table property: `isSupportMuslimCountSwitch`

Method Description:

```
+(void)setMuslimCountSwitch:(UInt8)isOpen block:(void(^)(int code, id data))block  
  
+(void)getMuslimCountSwitch:(void(^)(int code, id data))block
```

Parameter Description:

parameter	type	illustrate	illustrate
isOpen	Int		0: Off 1: Open

Example of usage:

```
//设置  
[DHBLECommand setMuslimCountSwitch:1 block:^(int code, id _Nonnull data) {  
  
}];  
  
//获取  
[DHBLECommand getMuslimCountSwitch:^(int code, id _Nonnull data) {  
    if (code == 0){  
        Boolean tOpen = [data boolValue];  
        NSLog(@"getMuslimCountSwitch tOpen %d", tOpen);  
    }  
}];
```

3.2.1.15 Get and set heart rate/blood oxygen alarm configuration.

This function allows you to set heart rate and blood oxygen level notification and alarm data; the alarm notification will be sent via `BluetoothNotificationRingHealthOverAlert`.
Configuration table attribute: `isSupportHrSpO2Alert`

Method Description:

```
+(void)getHRAAlert:(void(^)(int code, id data))block  
  
+(void)setHRAAlert:(DHHRAlertModel *)overModel block:(void(^)(int code, id data))block  
  
+(void)getSP02Alert:(void(^)(int code, id data))block  
  
+(void)setSP02Alert:(DHHRAlertModel *)overModel block:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	type	illustrate	illustrate
DHHRAlertModel	class		isOpen: YES (On), NO (Off); overValue: Alarm threshold, default values are heart rate exceeding 160 and blood oxygen below 94%. underValue: An alarm will be triggered if the value is below the set threshold; if the retrieved value is 0xff, it indicates that this function is not supported.

Note: If the `underValue` returned by `getHRAAlert()` is 0xff, it means this feature is not supported.

Example of usage:

```
//set
DHHRAlertModel *tHRAAlertModel = [[DHHRAlertModel alloc] init];
tHRAAlertModel.isOpen = YES;
tHRAAlertModel.overValue = 160;
tHRAAlertModel.underValue = 0xff;
[DHBLECommand setHRAAlert:tHRAAlertModel block:^(int code, id _Nonnull data) {

}];

//get
[DHBLECommand getHRAAlert:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHRAlertModel *model = data;
        NSLog(@"getHRAAlert %d %zd", model.isOpen, model.overValue);
    }
}];

// Alarm notification push
[[NSNotificationCenter defaultCenter] addObserver:self
selector:@selector(healthOverAlert:)
name:BluetoothNotificationRingHealthOverAlert object:nil];

- (void)healthOverAlert:(NSNotification *)ntf
{
    NSDictionary *tUserInfo = ntf.userInfo;
    NSInteger tType = [tUserInfo[@"type"] integerValue];
    NSInteger tValue = [tUserInfo[@"value"] integerValue];

    if (tType == 0){ //HeartRate Alert

    }
    else if (tType == 1){ //SP02

    }
}
```



```
}
```

3.2.1.16 Get and set screen timeout duration.

Configuration table property: `isBackLight`

Method Description:

```
+(void)getBrightTime:(void(^)(int code, id data))block
```

```
+(void)setBrightTime:(DHBrightTimeSetModel *)model block:(void(^)(int code, id data))block
```

Parameter Description:

DHBrightTimeSetModel Parameter	Type	Description	Value
duration	Int	Screen-on duration, in seconds (s), range 0-30s;	
durationNums	String	Supported duration values by the device; if available, separated by commas;	

3.2.1.17 Get and Set Wrist-Raise Screen-On Duration

Configuration table property: `isSupportRaisescreen`

Method Description:

```
+(void)ringGetGesture:(void(^)(int code, id data))block;
```

```
+(void)ringSetGesture:(DHGestureSetModel *)model block:(void(^)(int code, id data))block;
```

Parameter Description:

BrightScreenBean Parameter	Type	Description	Value
isOpen	Int	true: enabled; false: disabled	
startHour	Int	Start time (hour)	
startMin	Int	Start time (minute)	
endHour	Int	End time (hour)	
endMin	Int	End time (minute)	

3.2.1.18 Set Time Format (12-Hour / 24-Hour)

This setting only applies to devices with a display.

Method Description:

```
+(void)ringSetTimeformat:(UInt8)timeformat block:(void(^)(int code, id data))block;
```

Parameter Description:

Parameter	Type	Description	
timeformat	Int	0: 24-Hour 1: 12-Hour	

3.2.2 Health data synchronization (real-time single measurement and all-day monitoring)

There are two ways to monitor health data: real-time single measurements and continuous 24-hour monitoring. Health data includes heart rate, blood oxygen, stress levels, HRV, and sleep, **but sleep is not monitored in real time.**

(1) Real-time single detection: The app initiates a single detection on the device, and the results are returned immediately after the detection is complete.

(2) Continuous monitoring: You can set the interval time, for example, 30 minutes or 60 minutes, and the device will perform measurements and save the data; **if the app is not synchronized, the device can store 3-6 days of data.**

3.2.2.1 Real-time monitoring - Start and stop device health data monitoring.

Start health data monitoring (heart rate, blood oxygen, HRV, stress, blood sugar);

After the test is completed, the device notifies the app via the

`BluetoothNotificationHealthRingMeasureStateChange` notification;

The real-time test values are notified to the app via the

`BluetoothNotificationHealthRingMeasureValueChange` notification;

Method Description:

```
+(void)controlOpen:(NSInteger)type dataType:(NSInteger)dataType block:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type	illustrate	illustrate
dataType	NSInteger	Health data types	Heart Rate: BLE_KEY_HEART_RATE Blood Oxygen: BLEKEY_BLOOD_OXYGEN HRV: BLE_KEY_HRV Stress: BLE_KEY_STRESS Blood Sugar: BLE_KEY_BLOOD_SUGAR
type	NSInteger	Start/Stop	Start: 1 Stop: 0

Example of usage:

```

+ (void)controlOpen:(NSInteger)type dataType:(NSInteger)dataType block:(void(^)(
(int code, id data))block
//启动心率测试
[DHBleCommand controlOpen:1 dataType:BLE_KEY_HEART_RATE block:^(int code, id
_Nonnull data) {

}]];

//关闭心率测试
[DHBleCommand controlOpen:0 dataType:BLE_KEY_HEART_RATE block:^(int code, id
_Nonnull data) {

}]];

// 监听测量中实时数值改变
[[NSNotificationCenter defaultCenter] addObserver:self
selector:@selector(updateRingMeasureValueChange:)
name:BluetoothNotificationHealthRingMeasureValueChange object:nil];

- (void)updateRingMeasureValueChange:(NSNotification *)ntf
{
    NSDictionary *tUserInfo = ntf.userInfo;
    NSInteger tDataValue = [tUserInfo[@"dataValue"] integerValue];
    NSInteger tDataType = [tUserInfo[@"dataType"] integerValue];
    NSInteger tTimestamp = [tUserInfo[@"timestamp"] integerValue];

    NSLog(@"updateRingMeasureValueChange %04X 数值: %zd timestamp: %ld",
tDataType, tDataValue, tTimestamp);
    if (tDataType == BLE_KEY_APP_REAL_TIME_MUSLIM_COUNT){ //Muslim Count

    }
    else if (tDataType == BLE_KEY_APP_REAL_BLOOD_SUGAR_DATA){ //BloodSugar 血糖

    }
    else if (tDataType == BLE_KEY_APP_REAL_TIME_HRV_DATA){ //HRV 心率变异性

    }
    else if (tDataType == BLE_KEY_APP_REAL_TIME_HR_DATA){ //HR 心率

    }
    else if (tDataType == BLE_KEY_APP_REAL_TIME_BLOOD_OXYGEN_DATA){ //BloodOxygen
血氧

    }
    else if (tDataType == BLE_KEY_APP_REAL_TIME_STRESS_DATA){ //Stress 压力

    }
}
}

```

3.2.2.2 Continuous monitoring - Set the interval for continuous monitoring of health data.

Set the monitoring interval for health data (heart rate, blood oxygen, HRV, stress, blood glucose) throughout the day, in minutes.

Notes: Currently, only the heart rate interval can be set to 30 minutes or 60 minutes; other parameters (blood oxygen, HRV, stress, blood glucose) can only be set to on or off. The start and end times are fixed to cover the entire day and cannot be modified.

3.2.2.2.1 Heart rate detection settings and retrieval

The only interval options available for heart rate monitoring are 30 minutes and 60 minutes;

Method Description:

```
+(void)setHeartRateMode:(DHHeartRateModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getHeartRateMode:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type		
model	DHHeartRateModeSetModel	class	isOpen: true (on)/false (off) interval: interval time 30 or 60 minutes startHour: 0 (fixed, cannot be modified) startMin: 0 (fixed, cannot be modified) endHour: 23 (fixed, cannot be modified) endMin: 59 (fixed, cannot be modified);

Example of usage:

```
// 1. Set HeartRate Monitor(设置心率监听)
DHHeartRateModeSetModel *tModeSetModel = [[DHHeartRateModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 30; //可设置30分钟或60分钟
[DHBLECommand setHeartRateMode:tModeSetModel block:^(int code, id _Nonnull data)
{
    if (code == 0){
        NSLog(@"setHeartRateMode OK");
    }
}];

//1. 获取心率监听
[DHBLECommand getHeartRateMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHeartRateModeSetModel *model = data;
        NSLog(@"getHeartRateMode OK 开关 %d 检测周期 %d", model.isOpen,
model.interval);
    }
}];
```

3.2.2.2.2 Blood oxygen monitoring settings and data retrieval

The interval for blood oxygen measurements can only be set to 60 minutes;

Method Description:

```
+(void)setBoMode:(DHBoModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getBoMode:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type		
model	DHBoModeSetModel	Class	isOpen: true (on)/false (off) interval: Fixed interval of 60 minutes startHour: 0 (fixed, cannot be modified) startMin: 0 (fixed, cannot be modified) endHour: 23 (fixed, cannot be modified) endMin: 59 (fixed, cannot be modified);

Example of usage:

```
// 2. Set Blood oxygen Monitor(设置血氧监听)
DHBoModeSetModel *tModeSetModel = [[DHBoModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; //固定不可设置
[DHBLECommand setBoMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setBoMode OK");
    }
}];

//2. 获取血氧监听
[DHBLECommand getBoMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHBoModeSetModel *model = data;
        NSLog(@"getBoMode OK 开关 %d 检测周期 %d", model.isOpen, model.interval);
    }
}];
```

3.2.2.2.3 Heart Rate Variability (HRV) Measurement Settings and Data Acquisition

The HRV interval can only be set to 60 minutes;

Method Description:

```
+(void)setHrvMode:(DHHrvModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getHrvMode:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type		
model	DHHrvModeSetModel	Class	isOpen: true (on)/false (off) interval: Fixed interval of 60 minutes startHour: 0 (fixed, cannot be modified) startMin: 0 (fixed, cannot be modified) endHour: 23 (fixed, cannot be modified) endMin: 59 (fixed, cannot be modified);

Example of usage:

```
// 3. Set HRV Monitor(设置HRV监听)
DHHrvModeSetModel *tModeSetModel = [[DHHrvModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; //开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; //结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; //固定不可设置
[DHBleCommand setHrvMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0){
        NSLog(@"setHrvMode OK");
    }
}];

//3. 获取HRV监听
[DHBleCommand getHrvMode:^(int code, id _Nonnull data) {
    if (code == 0){
        DHHrvModeSetModel *model = data;
        NSLog(@"getHrvMode OK 开关 %d", model.isOpen);
    }
}];
```

3.2.2.2.4 Stress detection settings and retrieval

The interval stress can only be set to 60 minutes;

Method Description:

```
+(void)setStressMode:(DHStressModeSetModel *)model block:(void(^)(int code, id data))block

+(void)getStressMode:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type		
model	DHStressModeSetModel	Class	isOpen: true (on)/false (off) interval: Fixed interval of 60 minutes startHour: 0 (fixed, cannot be modified) startMin: 0 (fixed, cannot be modified) endHour: 23 (fixed, cannot be modified) endMin: 59 (fixed, cannot be modified);

Example of usage:

```
// 4. 设置压力监听
DHStressModeSetModel *tModeSetModel = [[DHStressModeSetModel alloc] init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; // 开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; // 结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; // 固定不可设置
[DHBLECommand setStressMode:tModeSetModel block:^(int code, id _Nonnull data) {
    if (code == 0) {
        NSLog(@"setStressMode OK");
    }
}]];

// 4. 获取压力监听
[DHBLECommand getStressMode:^(int code, id _Nonnull data) {
    if (code == 0) {
        DHStressModeSetModel *model = data;
        NSLog(@"getStressMode OK 开关 %d", model.isOpen);
    }
}]];

```

3.2.2.2.5 Blood glucose monitoring settings and data retrieval

The interval between blood glucose measurements can only be set to 60 minutes;

Method Description:

```
+(void)setBloodSugarMode:(DHBloodSugarModeSetModel *)model block:(void(^)(int code, id data))block
```

```
+(void)getBloodSugarMode:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type		
model	DHBloodSugarModeSetModel	Class	isOpen: true (on)/false (off) interval: Fixed interval of 60 minutes startHour: 0 (fixed, cannot be modified) startMin: 0 (fixed, cannot be modified) endHour: 23 (fixed, cannot be modified) endMin: 59 (fixed, cannot be modified);

Example of usage:

```
// 5. 设置血糖监听
DHBloodSugarModeSetModel *tModeSetModel = [[DHBloodSugarModeSetModel alloc]
init];
tModeSetModel.isOpen = YES;
tModeSetModel.startHour = 00; // 开始时间固定
tModeSetModel.startMinute = 00;
tModeSetModel.endHour = 23; // 结束时间固定
tModeSetModel.endMinute = 59;
tModeSetModel.interval = 60; // 固定不可设置
[DHBleCommand setBloodSugarMode:tModeSetModel block:^(int code, id _Nonnull
data) {
    if (code == 0) {
        NSLog(@"setBloodSugarMode OK");
    }
}]];

// 5. 获取血糖监听
[DHBleCommand getBloodSugarMode:^(int code, id _Nonnull data) {
    if (code == 0) {
        DHBloodSugarModeSetModel *model = data;
        NSLog(@"getBloodSugarMode OK 开关 %d", model.isOpen);
    }
}]];

```

3.2.2.3 24/7 monitoring - Synchronized health history data

Synchronizing health history data will automatically sync the corresponding health data based on the device's capabilities.

`data` is an array containing multi-day data of a specific type. It will return data of that type sequentially; `data` will not contain data of multiple types.

```
[DHBleCommand startDataSyncing:^(int code, id data){
    NSLog(@"同步完成 %d", code);
} datablock:^(int code, int progress, id _Nonnull data) {
    if (code == 0) {
        if ([data isKindOfClass:[NSArray class]]) {
            NSArray *array = data;
            for (id model in array) {

```



```

        if ([model isKindOfClass:[DHDailyStepModel class]]) {
            //Step
            NSLog(@"同步有 计步数据");
        }
        else if ([model isKindOfClass:[DHDailySleepModel
class]]) { //Sleep
            NSLog(@"同步有 睡眠数据");
        }
        else if ([model isKindOfClass:[DHDailyHrModel
class]]) { //HeartRate
            NSLog(@"同步有 心率数据");
        }
        else if ([model isKindOfClass:[DHDailyBoModel
class]]) { //BO
            NSLog(@"同步有 血氧数据");
        }
        else if ([model isKindOfClass:[DHDailyHrvModel
class]]) { ///HRV
            NSLog(@"同步有 HRV数据");
        }
        else if ([model isKindOfClass:[DHDailyPressureModel
class]]) { ///Stress
            NSLog(@"同步有 Stress数据");
        }
        else if ([model isKindOfClass:[DHDailyBloodSugarModel
class]]) { ///BloodSugar
            NSLog(@"同步有 血糖数据");
        }
        else if ([model isKindOfClass:
[DHDailyMuslimCountModel class]]) { ///Muslim count
            NSLog(@"同步有 赞念数据");
        }
    }
}
}
}];

```

3.2.2.4 24/7 Monitoring - Health Data Explanation

1. Step count data - DHDailyStepModel

```

@interface DHDailyStepModel : NSObject

/// 日期时间戳 (秒)
@property (nonatomic, copy) NSString *timestamp;
/// 日期yyyyMMdd
@property (nonatomic, copy) NSString *date;

/// 里程 (米)
@property (nonatomic, assign) NSInteger distance; //当天的总距离
/// 消耗 (卡路里)
@property (nonatomic, assign) NSInteger calorie; //当天总卡路里
/// 步数 (步)

```

```

@property (nonatomic, assign) NSInteger step; //当天总步数

/// 计步项 例:
@[@"index":@0,@"step":@100,@"calorie":@10000,@"distance":@50},...]
/// index (序号) step (步数) calorie (消耗) distance (里程) 单位同上
/// index从0开始, 固定24个item, 对应一天24小时
@property (nonatomic,strong) NSArray <NSDictionary *>*items; //一天时每小时的数据,总24条(今天未满足24条)

@end

```

2. Sleep data - DHDailySleepModel

```

@interface DHDailySleepModel : NSObject

/// 时间戳 (秒)
@property (nonatomic, copy) NSString *timestamp;
/// 日期yyyyMMdd
@property (nonatomic, copy) NSString *date;

/// 总时长 (分钟) Total duration (minutes)
@property (nonatomic, assign) NSInteger duration;
/// 入睡时间 (时间戳 (秒)) Time of falling asleep (timestamp in seconds)
@property (nonatomic, copy) NSString *beginTime;
/// 醒来时间 (时间戳 (秒)) Wake-up time (timestamp in seconds)
@property (nonatomic, copy) NSString *endTime;

/// 睡眠项 例: @[@"status":@0,@"value":@60},...]
/// status (睡眠类型) value (时长 (分钟))
/// status (0.awake 1. light sleep 2. deep sleep 3. REM)
@property (nonatomic,strong) NSMutableArray <NSDictionary *>*items;

@end

```

3. Heart rate data - DHDailyHrModel;

```

@interface DHDailyHrModel : NSObject

/// 时间戳 (秒)
@property (nonatomic, copy) NSString *timestamp;
/// 日期yyyyMMdd
@property (nonatomic, copy) NSString *date;

/// 心率项 例: @[@"timestamp":@0,@"value":@80},...]
/// timestamp (时间戳 (秒)) value (心率值)
@property (nonatomic,strong) NSMutableArray <NSDictionary *>*items;

@end

```

HRV `DHDailyHrvModel`, 压力 `DHDailyPressureModel`, 血氧 `DHDailyBoModel`, 血糖 `DHDailyBloodSugarModel` 与心率类似可参考对应类

4. Muslim prayer data - `DHDailyMuslimCountModel`

```
@interface DHDailyMuslimCountModel : NSObject
/// 时间戳 (秒)
@property (nonatomic, copy) NSString *timestamp;
/// 日期yyyyMMdd
@property (nonatomic, copy) NSString *date;

@property (nonatomic, assign) NSInteger muslimcount;

/// 赞念项 例: @[ @{@"timestamp":@0,@"value":@80},...]
/// timestamp (时间戳 (秒)) value (每小时赞念值)
@property (nonatomic, strong) NSMutableArray <NSDictionary *>*items; //注意,
这里的每小时为累加值赞念

@end
```

3.2.3 OTA upgrade

④ Note

The OTA update file must be obtained from the manufacturer and tested thoroughly before proceeding with the update. This is to prevent update errors and device malfunction.

```
[DHBLECommand startPXIUFileSyncing:[NSData dataWithContentsOfFile:tFilePath]
bleKey:BLE_KEY_ALL_BIN_DATA_PXI block:^(int code, CGFloat progress, id _Nonnull
data) {
    NSLog(@"code %d progress %f data %@", code, progress, data);
}];
```

3.2.4 Exercise more

⚠ Caution

The multi-sport configuration table property is `isSupportWorkout3`. After enabling multi-sport mode, the device will enter exercise mode. Neither disconnecting the app nor closing it will stop the activity; it can only be stopped manually through the app or the device itself. Therefore, for devices with multi-sport functionality, please check the status after connecting to determine if it is currently in exercise mode, as this may affect the use of other functions.

3.2.4.1 Get the device's multiple motion states.

Check if the device is currently engaged in multiple activities; only start a new activity if it's not currently engaged in any activity.

Method Description:

```
+(void)getControlSportWithRing:(void(^)(int code, id data))block
```

Parameter Description:

WorkoutControlType	Type		
Workout_Begin	Int	整形	Begin
Workout_Continue	Int	整形	Continue
Workout_Pause	Int	整形	Pause
Workout_Finish	Int	整形	Finish

Example of usage:

```
[DHBleCommand getControlSportWithRing:^(int code, id _Nonnull data) {  
    // @{@"keySportType":@(tSportType), @"keyControlType":@(tControlType)}  
    if (code == 0 && [data isKindOfClass:[NSDictionary class]]){  
        NSDictionary *tDic = data;  
        NSInteger tSportType = [tDic[@"keySportType"] integerValue];  
        WorkoutControlType tControlType = [tDic[@"keyControlType"] integerValue];  
    }  
}];
```

3.2.4.2 The control device enters multi-motion mode.

The control device enters multi-motion mode and starts the motion.

Changes in exercise data are obtained by receiving the

`BluetoothNotificationRingRuningData` notification.

Method Description:

```
+(void)controlSportWithRing:(DHSportControlModel *)model block:(void(^)(int code,  
id data))block
```

Parameter Description:

Parameter	Type		
DHSportControlModel	Class		controlType: refer to WorkoutControlType sportType: refer to BleActivityMode

Explanation of the data returned in the notification of changes in exercise data:

Parameter	Type		
ActivityTime	Int	整形	Duration of exercise, in seconds (s);
ActivitySteps	Int	整形	Steps taken during exercise
ActivityDistance	Int	整形	Distance is generated during movement, measured in meters (m);
ActivityCalorie	Int	整形	Heat is generated during exercise, measured in calories (cal);
ActivityHr	Int	整形	Dynamic heart rate during exercise
ActivityDataType	Int	整形	The source type, which is also returned by <code>setRingEnterWorkOut</code> .

Example of usage:

```
DHSportControlModel *model = [[DHSportControlModel alloc] init];
model.controlType = Workout_Begin; //开始
model.sportType = tbleActivityMode;
[DHBLECommand controlSportWithRing:model block:^(int code, id _Nonnull data) {
    if (code == 0){
        WorkoutRunningController *runningC = [[WorkoutRunningController alloc]
initWithNibName:@"WorkoutRunningController" bundle:nil];
        runningC.bleActivityMode = tbleActivityMode;
        runningC.controllType = Workout_Begin; //开始
        runningC.modalPresentationStyle = UIModalPresentationOverFullScreen;
        [weakSelf presentViewController:runningC animated:YES completion:^(
        });
    }
}];

[[NSNotificationCenter defaultCenter] addObserver:self
selector:@selector(ringRuningDataUdpate:)
name:BluetoothNotificationRingRuningData object:nil];

- (void)ringRuningDataUdpate:(NSNotification *)ntf
{
    NSDictionary *tUserInfo = ntf.userInfo;
    if (tUserInfo.count > 0){
        NSInteger duration = [tUserInfo[@"ActivityTime"] integerValue];
        NSInteger totalStep = [tUserInfo[@"ActivitySteps"] integerValue];
        NSInteger activityDistance = [tUserInfo[@"ActivityDistance"]
integerValue];
        NSInteger activityCalorie = [tUserInfo[@"ActivityCalorie"] integerValue];
```

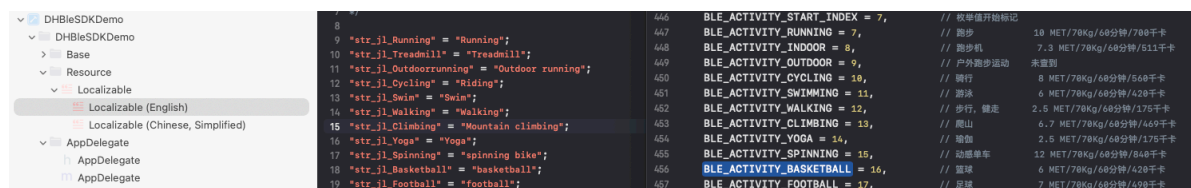
```

    NSInteger activityHr = [tUserInfo[@"ActivityHr"] integerValue];
    NSInteger tActivityDataType = [tUserInfo[@"ActivityDataType"]
integerValue];

    NSLog(@"ringRuningDataUdpate tActivityDataType %04X",
(UInt32)tActivityDataType);
    NSInteger hours = duration / 3600;
    NSInteger minutes = (duration % 3600) / 60;
    NSInteger seconds = duration % 60;
    self.workoutTimeLb.text = [NSString stringWithFormat:@"Time:
%02ld:%02ld:%02ld", (long)hours, (long)minutes, (long)seconds];
    self.workoutStepLb.text = [NSString stringWithFormat:@"Steps: %ld",
totalStep];
    self.workoutDistanceLb.text = [NSString stringWithFormat:@"Distance: %.2f
Km", floor(activityDistance/1000.0*100)/100];
    self.workoutCaloriesLb.text = [NSString stringWithFormat:@"Calories: %.1f
KCal", floor(activityCalorie/1000.0*10)/10];
    self.workoutHeartLb.text = [NSString stringWithFormat:@"HeartRate: %ld
bpm", activityHr];
    }
}

```

The corresponding names for BleActivityMode can be found in the example Demo string definitions:



3.2.4.3 Control enabling/disabling real-time notifications of motion data from the device.

控制开启/关闭设备实时通知运动数据;

运动中数据变化通过接收 `BluetoothNotificationRingRuningData` 通知获取,有时app关闭与进入后台,可告诉设备停止通知数据.

Method Description:

```
+(void)setRingEnterWorkOut:(UInt8)isEnter block:(void(^)(int code, id data))block
```

Parameter Description:

Parameter	Type	
isEnter	Int	1: Enable notification of exercise data; 0: Disable notification of exercise data;

Example of usage:

```
//退出运动界面
[DHBluetooth setRingEnterWorkOut:0 block:^(int code, id _Nonnull data) {

}];
```

3.2.4.4 Obtain multi-sport data reports.

Method Description:

```
+(void)startRingWorkout3Syncing:(void(^)(int code, id data))block dataBlock:
(void(^)(int code, int progress, id data))dataBlock
```

Return data DHDailySportModel parameter description:

DHDailySportModel 类	Type	
timestamp	NSString	Exercise start timestamp
date	NSString	Date-yyyyMMdd
viewType	Int	Current exercise types include: steps, cadence, pace, and distance: With cadence: viewTypeHaveStepFaq: Without step count: viewTypeNoStepNum: With pace: viewTypeHavePace: Without distance: viewTypeNoDistance:
heartRateItems	Array	The current list of heart rates generated during exercise, saved every minute;
.....		Other attributes can be found in the class documentation.

Example of usage:

```
[DHBluetooth startRingWorkout3Syncing:^(int code, id _Nonnull data) {
    NSLog(@"startRingWorkout3Syncing 同步完成 code %d", code);
} dataBlock:^(int code, int progress, id _Nonnull data) {
    if (code == 0) {
        if ([data isKindOfClass:[NSArray class]]){
            NSArray *array = data;
            for (id model in array) {
                if ([model isKindOfClass:[DHDailySportModel class]]) {
                    //保存数据库或其它操作
                }
            }
        }

        NSLog(@"startRingWorkout3Syncing data %zd", array.count);
    }
}
```

```
}  
}  
}];
```

SDK Revision History

.....

联系方式 / 技术支持

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